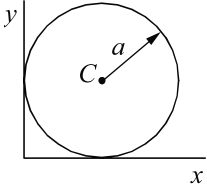
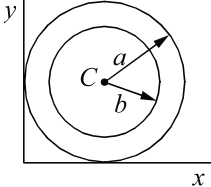
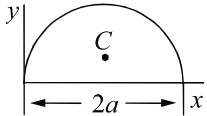
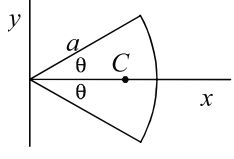
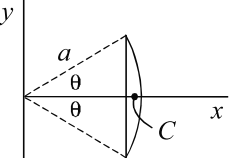


Figure	Area & Centroid	Area Moment of Inertia	(Radius of Gyration) ²	Product of Inertia
	$A = bh/2$ $x_c = 2b/3$ $y_c = h/3$	$I_{x_c} = bh^3/36$ $I_{y_c} = b^3h/36$ $I_x = bh^3/12$ $I_y = b^3h/4$	$r_{x_c}^2 = h^2/18$ $r_{y_c}^2 = b^2/18$ $r_x^2 = h^2/6$ $r_y^2 = b^2/2$	$I_{x_c y_c} = Abh/36 = b^2 h^2 / 72$ $I_{xy} = Abh/4 = b^2 h^2 / 8$
	$A = bh/2$ $x_c = b/3$ $y_c = h/3$	$I_{x_c} = bh^3/36$ $I_{y_c} = b^3h/36$ $I_x = bh^3/12$ $I_y = b^3h/12$	$r_{x_c}^2 = h^2/18$ $r_{y_c}^2 = b^2/18$ $r_x^2 = h^2/6$ $r_y^2 = b^2/6$	$I_{x_c y_c} = -Abh/36 = -b^2 h^2 / 72$ $I_{xy} = Abh/12 = b^2 h^2 / 24$
	$A = bh/2$ $x_c = (a+b)/3$ $y_c = h/3$	$I_{x_c} = bh^3/36$ $I_{y_c} = [bh(b^2 - ab + a^2)]/36$ $I_x = bh^3/12$ $I_y = [bh(b^2 + ab + a^2)]/12$	$r_{x_c}^2 = h^2/18$ $r_{y_c}^2 = (b^2 - ab + a^2)/18$ $r_x^2 = h^2/6$ $r_y^2 = (b^2 + ab + a^2)/6$	$I_{x_c y_c} = [Ah(2a-b)]/36$ $= [bh^2(2a-b)]/72$ $I_{xy} = [Ah(2a+b)]/12$ $= [bh^2(2a+b)]/24$
	$A = bh$ $x_c = b/2$ $y_c = h/2$	$I_{x_c} = bh^3/12$ $I_{y_c} = b^3h/12$ $I_x = bh^3/3$ $I_y = b^3h/3$ $J = [bh(b^2 + h^2)]/12$	$r_{x_c}^2 = h^2/12$ $r_{y_c}^2 = b^2/12$ $r_x^2 = h^2/3$ $r_y^2 = b^2/3$ $r_p^2 = (b^2 + h^2)/12$	$I_{x_c y_c} = 0$ $I_{xy} = Abh/4 = b^2 h^2 / 4$
	$A = h(a+b)/2$ $y_c = \frac{h(2a+b)}{3(a+b)}$	$I_{x_c} = \frac{h^3(a^2 + 4ab + b^2)}{36(a+b)}$ $I_x = \frac{h^3(3a+b)}{12}$	$r_{x_c}^2 = \frac{h^2(a^2 + 4ab + b^2)}{18(a+b)}$ $r_x^2 = \frac{h^2(3a+b)}{6(a+b)}$	
	$A = ab \sin \theta$ $x_c = (b + a \cos \theta)/2$ $y_c = (a \sin \theta)/2$	$I_{x_c} = (a^3 b \sin^3 \theta)/12$ $I_{y_c} = [ab \sin \theta (b^2 + a^2 \cos^2 \theta)]/12$ $I_x = (a^3 b \sin^3 \theta)/3$ $I_y = [ab \sin \theta (b + a \cos \theta)^2]/3$ $- (a^2 b^2 \sin \theta \cos \theta)/6$	$r_{x_c}^2 = (a \sin \theta)^2 / 12$ $r_{y_c}^2 = (b^2 + a^2 \cos^2 \theta) / 12$ $r_x^2 = (a \sin \theta)^2 / 3$ $r_y^2 = (b + a \cos \theta)^2 / 3$ $- (ab \cos \theta) / 6$	$I_{x_c y_c} = (a^3 b \sin^2 \theta \cos \theta) / 12$

Figure	Area & Centroid	Area Moment of Inertia	(Radius of Gyration) ²	Product of Inertia
	$A = \pi a^2$ $x_c = a$ $y_c = a$	$I_{x_c} = I_{y_c} = \pi a^4 / 4$ $I_x = I_y = 5\pi a^4 / 4$ $J = \pi a^4 / 2$	$r_{x_c}^2 = r_{y_c}^2 = a^2 / 4$ $r_x^2 = r_y^2 = 5a^2 / 4$ $r_p^2 = a^2 / 2$	$I_{x_c y_c} = 0$ $I_{xy} = A a^2$
	$A = \pi(a^2 - b^2)$ $x_c = a$ $y_c = a$	$I_{x_c} = I_{y_c} = \pi(a^4 - b^4) / 4$ $I_x = I_y = \frac{5\pi a^4}{4} - \pi a^2 b^2 - \frac{\pi b^4}{4}$ $J = \pi(a^4 - b^4) / 2$	$r_{x_c}^2 = r_{y_c}^2 = (a^2 + b^2) / 4$ $r_x^2 = r_y^2 = (5a^2 + b^2) / 4$ $r_p^2 = (a^2 + b^2) / 2$	$I_{x_c y_c} = 0$ $I_{xy} = A a^2$ $= \pi a^2 (a^2 - b^2)$
	$A = \pi a^2 / 2$ $x_c = a$ $y_c = 4a / (3\pi)$	$I_{x_c} = \frac{a^4(9\pi^2 - 64)}{72\pi}$ $I_{y_c} = \pi a^4 / 8$ $I_x = \pi a^4 / 8$ $I_y = 5\pi a^4 / 8$	$r_{x_c}^2 = \frac{a^2(9\pi^2 - 64)}{36\pi^2}$ $r_{y_c}^2 = a^2 / 4$ $r_x^2 = a^2 / 4$ $r_y^2 = 5a^2 / 4$	$I_{x_c y_c} = 0$ $I_{xy} = 2a^4 / 3$
 CIRCULAR SECTOR	$A = a^2 \theta$ $x_c = \frac{2a \sin \theta}{3 \theta}$ $y_c = 0$	$I_x = a^4 (\theta - \sin \theta \cos \theta) / 4$ $I_y = a^4 (\theta + \sin \theta \cos \theta) / 4$	$r_x^2 = \frac{a^2 (\theta - \sin \theta \cos \theta)}{4 \theta}$ $r_y^2 = \frac{a^2 (\theta + \sin \theta \cos \theta)}{4 \theta}$	$I_{x_c y_c} = 0$ $I_{xy} = 0$
 CIRCULAR SEGMENT	$A = a^2 \left[\theta - \frac{\sin 2\theta}{2} \right]$ $x_c = \frac{2a}{3} \frac{\sin^3 \theta}{\theta - \sin \theta \cos \theta}$ $y_c = 0$	$I_x = \frac{A a^2}{4} \left[1 - \frac{2 \sin^3 \theta \cos \theta}{3\theta - 3 \sin \theta \cos \theta} \right]$ $I_y = \frac{A a^2}{4} \left[1 + \frac{2 \sin^3 \theta \cos \theta}{\theta - \sin \theta \cos \theta} \right]$	$r_x^2 = \frac{a^2}{4} \left[1 - \frac{2 \sin^3 \theta \cos \theta}{3\theta - 3 \sin \theta \cos \theta} \right]$ $r_y^2 = \frac{a^2}{4} \left[1 + \frac{2 \sin^3 \theta \cos \theta}{\theta - \sin \theta \cos \theta} \right]$	$I_{x_c y_c} = 0$ $I_{xy} = 0$

$$Kn = \frac{\lambda}{L}$$

$$\tau = \mu \frac{du}{dy}$$

$$\tau = m \left| \frac{du}{dy} \right|^{n-1} \frac{du}{dy}$$

$$\mu = \mu_0 \left(\frac{1}{1 + \alpha t + \beta t^2} \right)$$

$$\mu = \mu_0 + \alpha t - \beta t^2$$

$$h = \frac{\sigma \times \pi d \times \cos \theta}{\frac{\pi}{4} d^2 \times \rho \times g} = \frac{4 \sigma \cos \theta}{\rho \times g \times d}$$

$$h^* = \frac{I_G \sin^2 \theta}{A \bar{h}} + \bar{h}$$

$$GM = \frac{I}{\nabla} - BG$$

$$T = 2\pi \sqrt{\frac{K^2}{GM \cdot g}} \quad I_{Y-Y} = \frac{W}{g} K^2$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0$$

$$\vec{\omega} = \frac{1}{2}(\nabla \times \vec{V})$$

$$\Gamma = \int_C V \cos \alpha \, ds$$

$$\zeta = \frac{\text{Circulation}}{\text{Area}}$$

$$= \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$\dot{\gamma}_{xy} = \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right)$$

$$\omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right)$$

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$u = -\frac{\partial \phi}{\partial x}$$

$$v = -\frac{\partial \phi}{\partial y}$$

$$w = -\frac{\partial \phi}{\partial z}$$

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z}$$

$$1 \text{ Bar} = 100000 \text{ N/m}^2$$

$$1 \text{ Bar} = 0.986923 \text{ atm}$$

$$1 \text{ atm} = 101325 \text{ Pa}$$

$$\frac{\partial \psi}{\partial x} = v$$

$$\frac{\partial \psi}{\partial y} = -u$$